Statistical Analysis of Global Temperature Trends from Historical Climate Observations Using Regression Modeling

A Quantitative Climate Data Science Analysis of Long-Term Temperature Anomalies

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Abstract

Understanding long-term global temperature change is a central problem in climate science because changes in global mean surface temperature influence sea level rise, ecosystem stability, agricultural productivity, and the frequency of climate extremes. This report presents a quantitative analysis of historical global temperature anomalies using statistical regression models applied to the Berkeley Earth global land/ocean temperature record. The objective was to build a reproducible workflow for processing climate observations, estimating long-term warming trends, comparing statistical model performance, and communicating results through scientific figures and an interactive dashboard.

Monthly observations were parsed from the original Berkeley Earth text file, cleaned, converted into numerical format, and aggregated into annual temperature anomaly values. Two regression models were compared: ordinary least squares Linear Regression and second-order Polynomial Regression. Model performance was evaluated using the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). A separate post-1970 regression analysis was used to estimate the contemporary warming rate.

The Polynomial Regression model achieved stronger explanatory performance ($R^2 = 0.8993$) than the Linear Regression model ($R^2 = 0.7372$), indicating that the historical global temperature record is better described by a nonlinear trend than by a single constant warming rate. The estimated warming rate since 1970 was approximately 0.194°C per decade. These results should be interpreted as statistical descriptions of historical observations rather than physical climate simulations. Overall, the project demonstrates how mathematical modeling, statistical evaluation, scientific programming, and climate data visualization can be combined into a reproducible environmental data science workflow.

Keywords: climate change; global temperature anomaly; Berkeley Earth; regression modeling; polynomial regression; environmental statistics; Python; climate data science

1 Introduction

Global surface temperature is one of the most widely used indicators of Earth's changing climate. Since the beginning of industrialization, rising greenhouse gas concentrations have contributed to measurable increases in global mean temperature. Even relatively small changes in average

temperature can affect weather patterns, sea level rise, hydrological cycles, agricultural systems, and ecological stability. Long-term temperature records therefore provide a fundamental basis for understanding climate change and communicating its magnitude.

Organizations such as Berkeley Earth, NASA, NOAA, and the Intergovernmental Panel on Climate Change maintain observational temperature datasets that allow researchers to examine changes across long time periods. These datasets combine land-based weather station observations and ocean temperature measurements to estimate global temperature anomalies. Temperature anomalies, rather than absolute temperatures, are commonly used because they measure departures from a reference climatology and reduce the influence of regional differences in baseline climate.

Physical climate models simulate the climate system through equations representing radiation, atmospheric circulation, ocean heat transport, land-atmosphere interactions, and other processes. Statistical models are different: they do not simulate the physical climate system directly, but they can identify patterns in observed data and quantify long-term trends. Regression analysis is one of the most fundamental tools for this type of analysis. Linear Regression estimates a constant rate of change, while Polynomial Regression allows curvature and can describe accelerating or decelerating trends.

The purpose of this project was to analyze historical global temperature anomalies using regression modeling. The project was designed as a technical portfolio report rather than a novel climate model. Its main goals were to demonstrate climate data preprocessing, statistical model construction, regression evaluation, scientific visualization, and transparent interpretation of results.

2 Research Question and Objectives

The central research question was:

> Can regression models be used to quantify and compare long-term global temperature anomaly trends from historical climate observations?

The specific objectives were:

1. Build a reproducible Python workflow for parsing and cleaning Berkeley Earth temperature anomaly data.
2. Aggregate monthly observations into annual global temperature anomaly values.
3. Compare Linear Regression and Polynomial Regression as statistical descriptions of historical warming.
4. Evaluate model performance using R^2, MAE, and RMSE.
5. Estimate the contemporary warming rate since 1970.
6. Generate figures and a Streamlit dashboard to communicate the analysis.

3 Dataset and Preprocessing

3.1 Dataset

This analysis used the Berkeley Earth Global Land and Ocean Temperature dataset, an open climate record that combines land-surface temperature information with ocean temperature data. The original text file includes metadata, citation information, and numerical observations. For this project, the numerical records were extracted and processed in Python.

The dataset reports temperature anomalies rather than absolute temperatures. A temperature anomaly represents the difference between an observed temperature and a reference baseline. This representation is especially useful for global analysis because it allows observations from different regions and seasons to be compared on a common scale.

3.2 Variables

The final modeling dataset used two primary variables:

Table 1: Variables used in the temperature trend models.

Variable	Description	Role
Year	Calendar year of observation	Predictor
Temperature anomaly (°C)	Annual global temperature anomaly	Response

For exploratory visualization, monthly anomaly values were also retained. Annual anomalies were created by averaging monthly anomalies within each calendar year.

3.3 Data Cleaning

The original Berkeley Earth file contains comment lines and descriptive text before the numerical data. These lines were skipped during parsing. The preprocessing workflow included:

- reading the text file line by line;
- extracting year, month, anomaly, and uncertainty values;
- filtering records to valid calendar years and months;
- converting numerical fields into Python numeric types;
- constructing monthly and annual data frames;
- aggregating monthly anomalies into annual means;
- exporting cleaned annual data for reproducibility.

This workflow produced a clean time series suitable for regression modeling and visualization.

4 Methodology

4.1 Analytical Workflow

The analysis followed a standard environmental data science workflow: data acquisition, parsing, cleaning, annual aggregation, exploratory visualization, model fitting, model evaluation, trend interpretation, and dashboard deployment. The same annual anomaly series was used for both regression models so that their performance could be compared directly.

4.2 Linear Regression

Linear Regression was used as a baseline model because it provides an interpretable estimate of average warming rate over time. The model can be written as

$$y = \beta_0 + \beta_1 x + \epsilon, \tag{1}$$

where y is annual temperature anomaly, x is year, β_0 is the intercept, β_1 is the estimated slope, and ϵ is the residual error. The slope β_1 represents the average annual rate of change under the assumption that warming occurs at a constant rate across the full record.

4.3 Polynomial Regression

Visual inspection of the temperature record suggested that warming has not followed a perfectly linear trajectory. To represent possible acceleration, a second-order Polynomial Regression model was also fitted:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon. \tag{2}$$

The quadratic term allows the fitted curve to bend over time. This makes the model more flexible than Linear Regression while still remaining relatively simple and interpretable. The polynomial model is a statistical trend model; it does not represent a physical climate mechanism by itself.

4.4 Model Evaluation Metrics

Model performance was evaluated using three regression metrics. The coefficient of determination is

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{3}$$

R^2 measures the proportion of observed variation explained by the model. MAE is

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|, \tag{4}$$

and RMSE is

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{5}$$

MAE measures average absolute prediction error, while RMSE penalizes larger errors more strongly.

4.5 Warming Rate Since 1970

To estimate the contemporary warming rate, a separate Linear Regression model was fitted using observations beginning in 1970. The estimated annual slope was converted into a decadal rate:

$$\text{Warming Rate} = 10 \times \beta_1. \tag{6}$$

This produced an estimated warming rate of approximately 0.194°C per decade.

5 Results

5.1 Model Comparison

The two regression models produced substantially different fits to the historical temperature record. The Linear Regression model explained approximately 73.7% of the observed variation in annual anomalies, while the Polynomial Regression model explained approximately 89.9%.

Table 2: Regression model performance for annual global temperature anomalies.

Model	R^2	**MAE**	**RMSE**
Linear Regression	0.7372	0.1649	0.2025
Quadratic Regression	**0.8993**	**0.0991**	**0.1254**

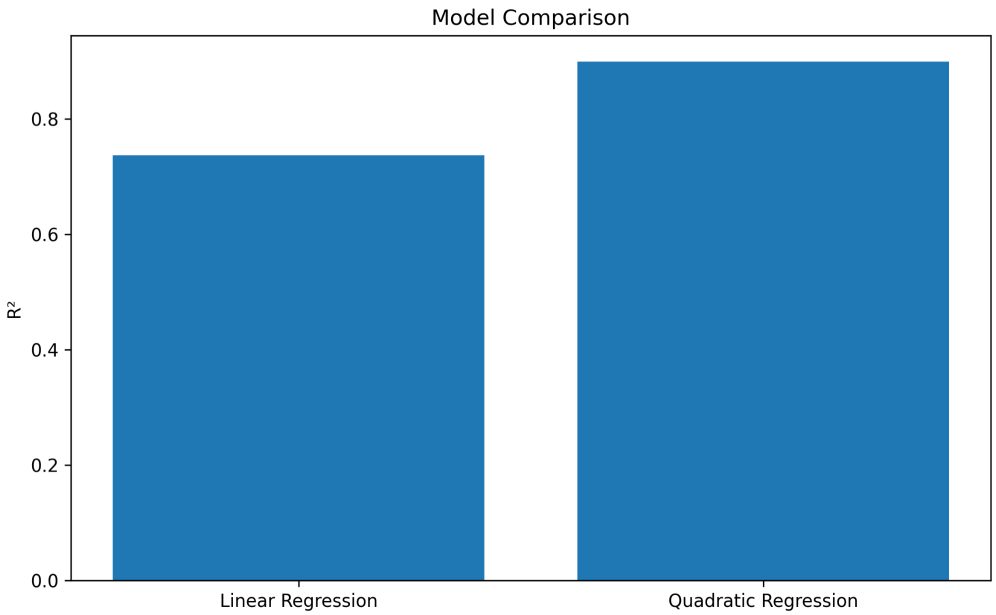


Figure 1: Comparison of model R^2 values. Quadratic Regression achieved the higher explanatory performance.

The higher R^2 and lower error values for Quadratic Regression indicate that a nonlinear trend better represents the historical anomaly series than a single straight line.

5.2 Annual Temperature Anomaly Trend

Figure 2 shows the annual global temperature anomaly record with fitted Linear and Quadratic Regression curves. The annual series shows substantial year-to-year variability, but the long-term warming signal is clear. The linear trend captures the broad upward direction, while the quadratic trend better follows the curvature of the observed record, especially during the late twentieth and early twenty-first centuries.

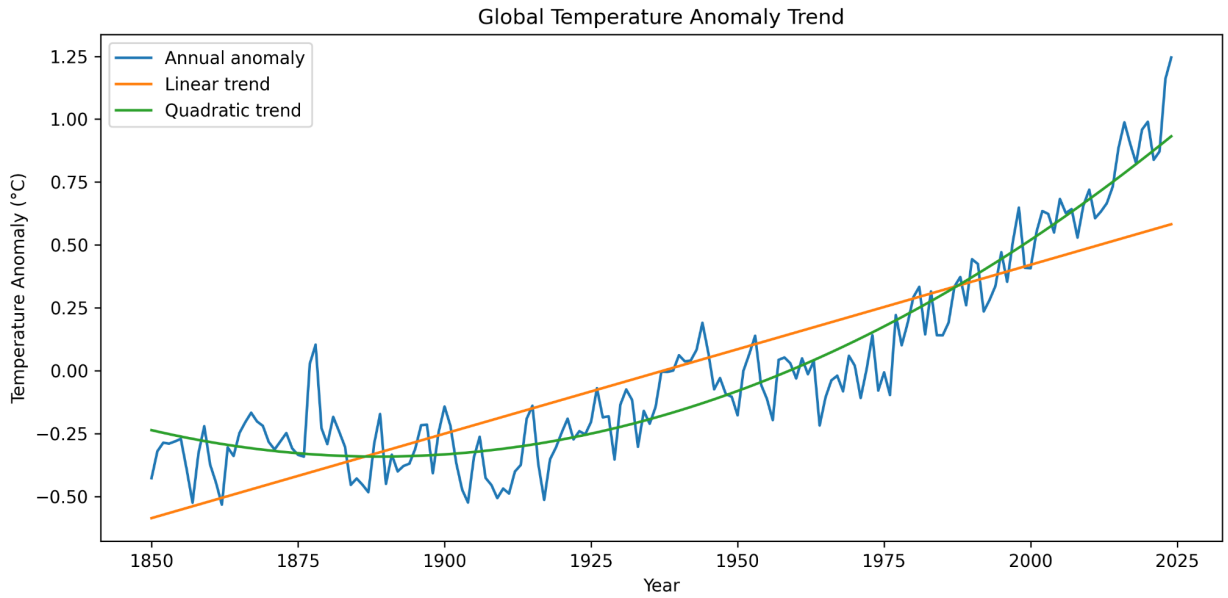


Figure 2: Annual global temperature anomalies with fitted Linear Regression and Quadratic Regression trends. The quadratic curve better captures the nonlinear acceleration in recent decades.

5.3 Monthly Temperature Anomalies

Monthly observations show greater short-term variability than annual averages, but the long-term upward trend remains visible. Figure 3 illustrates that monthly anomalies fluctuate around the broader warming trajectory, with recent decades showing consistently higher anomaly values than the nineteenth and early twentieth centuries.

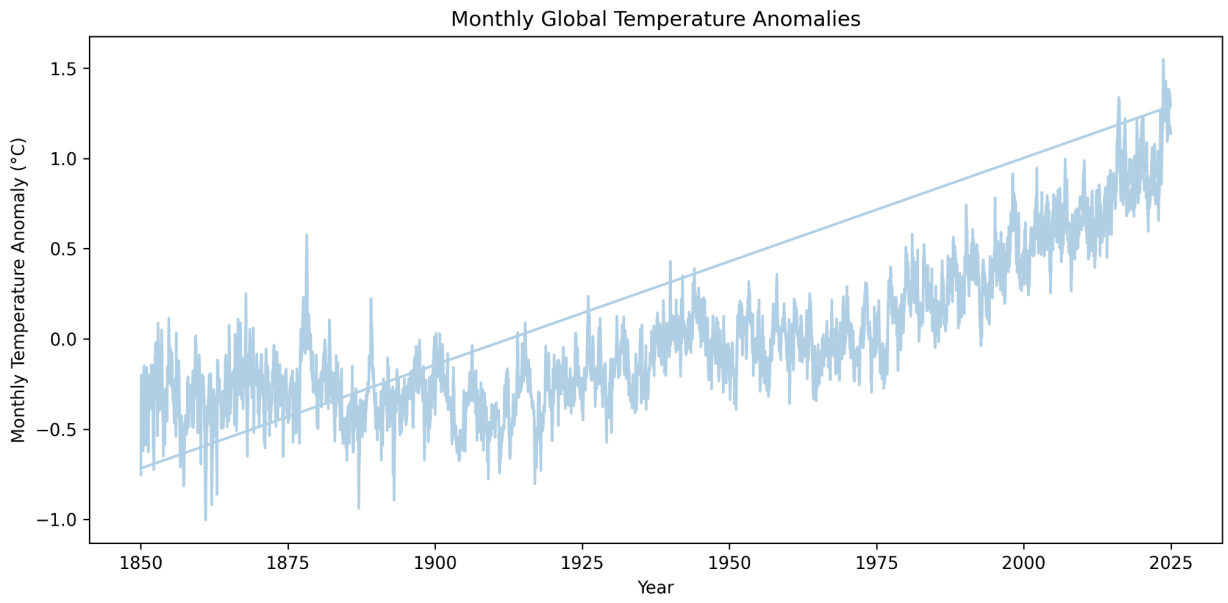


Figure 3: Monthly global temperature anomalies. Short-term variability is substantial, but the long-term warming trend is evident across the full record.

5.4 Contemporary Warming Rate

The post-1970 warming rate was estimated as approximately

$$\boxed{0.194^{\circ}\text{C per decade}}. \tag{7}$$

This value indicates that global average temperature anomalies increased by nearly two-tenths of a degree Celsius per decade during the modern warming period.

6 Discussion

6.1 Statistical Interpretation

The improvement from Linear Regression to Quadratic Regression shows that global temperature anomalies are not fully described by a constant rate of change over the entire observational period. A straight line captures the overall direction of warming, but it underrepresents the curvature visible in the record. The quadratic model adds flexibility through the x^2 term and therefore better approximates the historical trajectory.

This does not mean the quadratic model is a physical climate model. It is a descriptive statistical model. Its purpose is to summarize the shape of the observed time series, not to simulate greenhouse gas forcing, ocean heat uptake, atmospheric dynamics, or feedback mechanisms.

6.2 Climate Interpretation

The annual and monthly figures both show a persistent upward shift in global temperature anomalies. Early observations fluctuate largely below zero anomaly values, while recent observations are consistently positive and reach substantially higher values. The post-1970 warming rate estimate supports the interpretation that recent warming has been rapid relative to much of the earlier record.

6.3 Limitations

Several limitations should be considered. First, the analysis uses global average anomalies, which hide regional differences in climate change. Second, the models are statistical trend fits rather than mechanistic climate simulations. Third, Polynomial Regression can fit historical curvature but may produce unrealistic extrapolations if extended far beyond the observed data. Fourth, uncertainty in measurement, data coverage, and historical reconstruction was not explicitly propagated through the regression models.

6.4 Future Work

Future extensions could include time-series models such as ARIMA, state-space modeling, Bayesian regression, Gaussian Process Regression, and change-point detection. The analysis could also be expanded to regional temperature datasets, comparisons with CMIP6 model projections, or integration with other climate indicators such as atmospheric carbon dioxide, sea level rise, ocean heat content, or Arctic sea ice extent.

7 Conclusion

This project developed a reproducible statistical workflow for analyzing global temperature anomaly trends using the Berkeley Earth dataset. Linear and Quadratic Regression models were compared, and Quadratic Regression provided the stronger statistical representation of the historical record, with $R^2 = 0.8993$ compared with $R^2 = 0.7372$ for Linear Regression. The estimated warming rate since 1970 was approximately 0.194°C per decade.

The project demonstrates how mathematical modeling, statistical evaluation, Python programming, and climate visualization can be combined to analyze a real environmental dataset. While the analysis is descriptive rather than mechanistic, it provides a strong portfolio example of quantitative climate data science and creates a foundation for more advanced work in time-series analysis, uncertainty quantification, and climate model comparison.

Technical Skills Demonstrated

- **Programming:** Python, pandas, NumPy, scikit-learn, Matplotlib, Plotly, Streamlit.
- **Statistics and modeling:** Linear Regression, Polynomial Regression, residual error metrics, model comparison, trend estimation.

- **Climate data science:** temperature anomalies, annual aggregation, warming rate estimation, interpretation of global climate records.
- **Scientific communication:** reproducible figures, dashboard deployment, structured technical reporting, LaTeX manuscript formatting.

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